
KESTREL

The drone security analyst that never blinks

Feature Spec

What KESTREL is for, and the three requirements that define it

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Deliverable 1 of the assignment

Every figure in this document is generated from a measured run. Re-running the evaluation suite regenerates the tables; no number is transcribed by hand.

FlytBase AI Engineer assignment

1 Value to a site owner

A docked patrol drone monitoring a fixed site produces roughly **57,600 frames per eight-hour shift** at a modest 2 fps. Almost every one shows the same empty yard. The footage is recorded; almost none of it is watched. When something does happen it is discovered afterwards, by someone scrubbing video looking for a moment they already know occurred.

KESTREL is the analyst that watches all of it.

Continuous attention	Every frame is assessed. Not every frame is analysed — that is the point — but nothing is unwatched.
Alerts worth reading	Suppression, counterfactual checks and learned feedback exist so the alerts that arrive are worth an operator’s time. A system raising forty alerts a night is switched off within a week, and then protects nothing.
Memory across days	<i>“The same vehicle, seventh visit, and the first time it has ever been here after midnight.”</i> No single frame is alarming; only the pattern is.
Answers, not archives	Ask “what happened last night” and get a cited narrative, not a folder of MP4s.
Something to do	Every alert carries dispatch coordinates, bearing, ETA and a feasibility-checked flight plan. At 02:00 with one guard on shift, knowing <i>something happened</i> is the easy part.

2 R1 — Decide what deserves attention, under a hard cost ceiling

Requirement. Process a continuous frame stream within a fixed inference budget, and be able to state and defend how much of that stream was analysed.

Why it comes first. The free inference tier permits roughly 40 requests per minute. Captioning 57,600 frames would take about twenty-four hours of wall-clock and cost more than the guard it replaces. Any design treating “call a model on each frame” as the baseline is not deployable.

How. A tier-0 gate runs three cheap CPU signals — perceptual-hash distance, pixel delta, embedding novelty. Sensitivity adapts to context and is capped so the gate cannot be tuned into uselessness.

Acceptance. The skip rate is measured per context, and every decision *including the skips* is persisted so the figure is auditable.

38.9% of frames never reached a model on real footage, **96.7%** on an idle patrol

3 R2 — Understand and remember, not merely detect

Requirement. Produce structured, queryable observations and maintain identity for subjects across frames, sessions and days.

Why. “A person was detected” is not a security finding. “A person has been at the gate for four minutes, at 00:01, and this vehicle has visited seven times but never at this hour” is. The second sentence needs three things a detector cannot provide: structure, persistent identity, and a model of what is normal.

How. A structured scene graph rather than a caption; persistent entities matched by appearance, attributes and spatio-temporal plausibility; a salience-weighted temporal memory

pyramid; and a normalcy baseline that *abstains* below three days of history rather than flooding an operator with false novelty on day one.

Acceptance. Both outputs named in the brief are produced from the running system, and a subject’s visit history is queryable across days.

4 R3 — Be right often enough to stay trusted

Requirement. Alerts carry their evidence; suppress what does not warrant a human; refuse to answer without grounds; never take a consequential action without explicit approval.

Why. The failure mode that kills a security product is not a missed detection — it is a stream of confident wrong ones. Trust, once spent, is not recovered.

How. Evidence chains on every alert. Triage that can decline: contradicting evidence suppresses outright (a dog is not a person, and scaling a confidence cannot express “this premise is false”), while mitigating evidence scales — though PPE never excuses a critical-zone breach, because a hi-vis vest is not an access credential. Grounded answers, verified against what tools actually returned. And a permission boundary enforced in code rather than in prompt wording.

Scenario suite

8/8 scenarios correct 1.00 precision 1.00 recall 1.00 F_1

True-negative scenarios (routine delivery, wildlife at the fence, shift change) are weighted equally in this suite. A security system that cries wolf is switched off, and then protects nothing.

5 Scope

In scope. Frame and telemetry ingest; the perception cascade; entity memory and baselines; a temporal rule language with natural-language authoring and backtesting; hybrid retrieval; mission proposal with feasibility; a conversational control plane; a portfolio view; an audit ledger.

Out of scope, deliberately. Flight control; multi-camera fusion; face recognition or any biometric identification; live streaming protocols; accounts and role-based access.

Simulated, and labelled wherever it appears

Telemetry — there is no aircraft. Site geometry. Mission execution. Every fleet site except the flagship plant.

Real

The video, the detections, the tracks, the embeddings, the captions, the rules, the memory, the retrieval, the audit chain, and every measured number in this document.